

PROJECT 1

Creating a VM on Azure and Hosting a Static Website

Submitted By

Bhavani M Ghorpade

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ABSTRACT

Cloud computing has become one of the most important technologies in the modern IT industry. Instead of maintaining physical servers and hardware systems, organizations now use cloud platforms to store data, deploy applications, and manage computing resources through the internet. Microsoft Azure is one of the leading cloud platforms that provides scalable and secure cloud services for businesses, developers, and students.

This project focuses on creating a Linux Virtual Machine on Microsoft Azure and hosting a static website using Apache Web Server. The website was developed using HTML and CSS and deployed on Ubuntu Linux running inside the Azure cloud environment. The project demonstrates practical implementation of cloud infrastructure setup, server configuration, networking, remote access using SSH, and web hosting.

The implementation process involved creating an Azure Virtual Machine, configuring inbound security rules for SSH and HTTP communication, remotely connecting to the server using PowerShell, installing Apache2 web server, creating HTML webpage files, and verifying website deployment through a web browser using the public IP address.

This project helped in understanding real-world cloud deployment procedures and provided practical experience in Linux server administration, cloud computing concepts, and website hosting technologies.

PROBLEM STATEMENT

Traditional website hosting requires physical servers, infrastructure maintenance, and high setup costs. Small organizations, students, and startups often face challenges in deploying websites due to hardware limitations and management complexity. Cloud platforms such as Microsoft Azure provide an efficient solution by offering scalable virtual infrastructure that can host websites with minimal cost and maintenance effort.

OBJECTIVE

The objectives of this project are:

1. To understand the concept of cloud computing.
2. To learn the fundamentals of Microsoft Azure services.
3. To create and configure a Linux Virtual Machine.
4. To understand network security configuration.
5. To establish SSH-based remote server access.
6. To install and configure Apache2 Web Server.
7. To create a static website using HTML and CSS.
8. To deploy the website on Azure Virtual Machine.
9. To understand real-world cloud hosting architecture.
10. To gain hands-on experience in cloud deployment.

INTRODUCTION

Cloud Computing

Cloud computing refers to the delivery of computing services such as servers, databases, storage, software, and networking through the internet. It allows users to access resources on demand without maintaining physical hardware systems.

Today, cloud computing is used by companies, educational institutions, startups, and developers because it provides flexibility, scalability, and reduced operational cost.

In modern computing environments, cloud computing has become a backbone for application deployment and data management. One of the key reasons for its rapid adoption is that it removes dependency on physical infrastructure. Earlier, organizations had to purchase and maintain expensive servers, but now everything can be provisioned virtually within minutes using cloud platforms.

Another important aspect of cloud computing is its pay-as-you-go model, where users are charged only for the resources they consume. This makes it highly suitable for students, startups, and small organizations that cannot afford large infrastructure investments. Advantages of cloud computing include:

- Scalability
- Cost Efficiency
- Better Security
- Remote Accessibility
- High Availability
- Easy Resource Management

Cloud services are mainly divided into three categories:

1. Infrastructure as a Service (IaaS)
2. Platform as a Service (PaaS)
3. Software as a Service (SaaS)

This project uses Infrastructure as a Service through Microsoft Azure Virtual Machines.

Microsoft Azure

Microsoft Azure is a cloud computing platform developed by Microsoft. It provides services such as virtual machines, networking, databases, storage systems, AI services, analytics, and website hosting.

Azure allows developers and organizations to deploy applications globally with reliability and security. It supports both Linux and Windows operating systems.

Some major Azure services include:

- Azure Virtual Machines
- Azure Storage
- Azure Networking
- Azure App Services
- Azure AI Services
- Azure SQL Database

Microsoft Azure also provides a highly secure and scalable environment for deploying applications. One of its key strengths is global availability, meaning applications hosted on Azure can be accessed from different parts of the world with minimal latency.

In addition, Azure provides built-in monitoring and security tools that help track system performance and protect resources from unauthorized access. These features make Azure suitable not only for learning purposes but also for enterprise-level applications.

Virtual Machine

A Virtual Machine is a software-based computer system that behaves like a physical computer. It includes:

- Virtual CPU
- Virtual Memory
- Virtual Storage
- Operating System
- Networking Components

Virtual machines are widely used in cloud computing because they provide isolation between different systems running on the same physical hardware. Each virtual machine behaves like an independent computer with its own operating system, memory, and storage.

This isolation ensures that if one virtual machine fails, it does not affect others running on the same physical server. Because of this reliability, virtual machines are commonly used in production environments for hosting applications and services.[]

Applications of Virtual Machines include:

- Website Hosting
- Application Deployment
- Software Testing
- Cloud Infrastructure
- Remote Server Management

Static Website

A static website contains fixed webpages created using HTML and CSS. Static websites are simple, secure, and easy to deploy.

Advantages of static websites include:

- Faster Loading Speed
- Better Security
- Lower Hosting Cost
- Easy Deployment
- Simple Maintenance

Apache Web Server

Apache HTTP Server is one of the most popular open-source web servers. It is widely used to host websites and applications.

Features of Apache include:

- Open Source
- Reliable
- Secure

- Cross Platform
- Easy Configuration

Apache stores website files inside:

```
/var/www/html
```

FUNDAMENTALS OF CLOUD COMPUTING IN DETAIL

Cloud computing is a model that enables on-demand access to computing resources such as servers, storage, databases, and networking over the internet.

Deployment Models

- Public Cloud: Services offered over the public internet (Azure, AWS)
- Private Cloud: Dedicated infrastructure for a single organization
- Hybrid Cloud: Combination of public and private cloud

Service Models

- Infrastructure as a Service (IaaS): Virtual machines and networking (used in this project)
- Platform as a Service (PaaS): Application hosting platforms
- Software as a Service (SaaS): Ready-to-use software applications

Importance of Cloud Computing

Cloud computing reduces infrastructure cost, improves scalability, and enables remote access to computing resources from anywhere in the world.

SYSTEM REQUIREMENTS

Hardware Requirements

- Laptop/Desktop Computer
- Minimum 4GB RAM
- Stable Internet Connection

Software Requirements

- Microsoft Azure Account
- Ubuntu Server 22.04 LTS
- PowerShell or Command Prompt
- Apache2 Web Server
- Nano Text Editor
- Web Browser

ARCHITECTURE OF WEBSITE HOSTING

The architecture of this project consists of a user browser connected to the Azure Virtual Machine through the internet. The VM runs Ubuntu Linux and Apache Web Server which hosts the static website files.

User Browser
HTTP Request
Azure Public IP Address
Azure Virtual Machine
Ubuntu 22.04 LTS
Apache Web Server
HTML Website Files

MICROSOFT AZURE ARCHITECTURE

Microsoft Azure provides cloud infrastructure through globally distributed data centers.

Key Components Used in This Project

- Resource Group: Logical container for Azure resources
- Virtual Machine: Compute instance running Ubuntu Linux
- Virtual Network: Provides communication between resources
- Network Security Group (NSG): Controls inbound/outbound traffic
- Public IP Address: Enables internet access to VM

Networking Concept

The Virtual Machine is assigned a public IP address which allows external users to access the hosted website via HTTP protocol on port 80.

Explanation of Architecture Flow

The architecture of this project follows a simple client-server model. The user accesses the website through a web browser, which sends an HTTP request to the Azure Virtual Machine using its public IP address.

The request is received by the Apache Web Server running inside the Ubuntu operating system. The server then processes the request and returns the HTML webpage back to the user.

This entire process happens over the internet within a few milliseconds, making cloud-based hosting efficient and reliable.

METHODOLOGY

Step 1: Creating Azure Virtual Machine

1. Open Microsoft Azure Portal.
2. Navigate to Virtual Machines section.
3. Click Create → Azure Virtual Machine.
4. Enter Virtual Machine name.
5. Select Ubuntu Server 22.04 LTS.
6. Configure username and password.
7. Enable inbound ports 22 and 80.
8. Click Review + Create.
9. Deploy the Virtual Machine.

Virtual Machine Configuration

Parameter	Value
VM Name	Myvm
Operating System	Ubuntu 22.04 LTS
Authentication Type	Password
Username	azureuser
Ports Enabled	22 and 80

Step 2: SSH Connection

SSH was used to connect remotely to the Linux server.

```
ssh azureuser@20.193.153.54
```

Step 3: Updating Ubuntu Packages

```
sudo apt update
```

Step 4: Installing Apache Web Server

```
sudo apt install apache2 -y
```

Step 5: Navigating to Web Directory

```
cd /var/www/html
```

Step 6: Removing Default Webpage

```
sudo rm index.html
```

Step 7: Creating HTML File

```
sudo nano index.html
```

CHALLENGES FACED

During the implementation of this project, several practical challenges were encountered:

- Configuring inbound security rules correctly.
- Establishing SSH connectivity to the virtual machine.
- Managing Linux file permissions.
- Installing and verifying Apache services.
- Troubleshooting network accessibility issues.
- Ensuring website availability through the public IP address.

CODE

HTML CODE

```
<!DOCTYPE html>
<html>
<head>
<title>Azure Static Website</title>
</head>

<body style="font-family: Arial;
            text-align: center;
            margin-top: 100px;">

<h1>Success!</h1>

<p>This website is hosted on Azure
by Bhavani M Ghorpade.</p>

<p>Electronics and Communication
Engineering Project 2026.</p>

</body>
</html>
```

CSS Styling

```
body{
font-family: Arial;
text-align: center;
margin-top: 100px;
}
```

OUTPUT SCREENSHOTS

The output section demonstrates the successful execution of all implemented steps in this project. Each screenshot represents a different stage of the cloud deployment process, starting from virtual machine creation to final website hosting.

These outputs confirm that the system is working as expected and the cloud environment has been configured correctly. The final result clearly shows that the website is accessible through the public IP address of the Azure Virtual Machine.

Azure Virtual Machine Dashboard

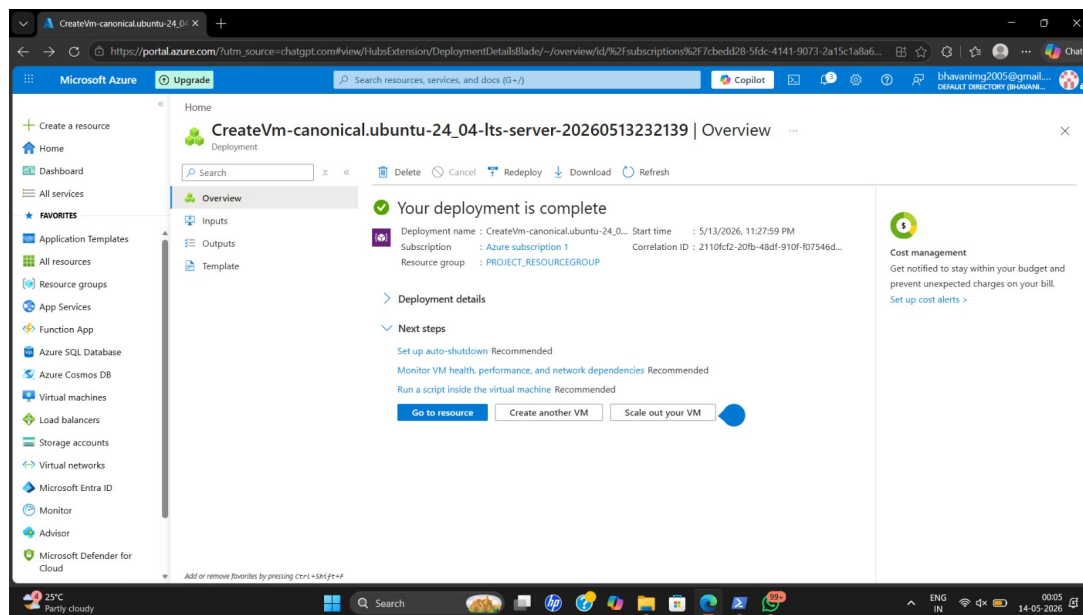
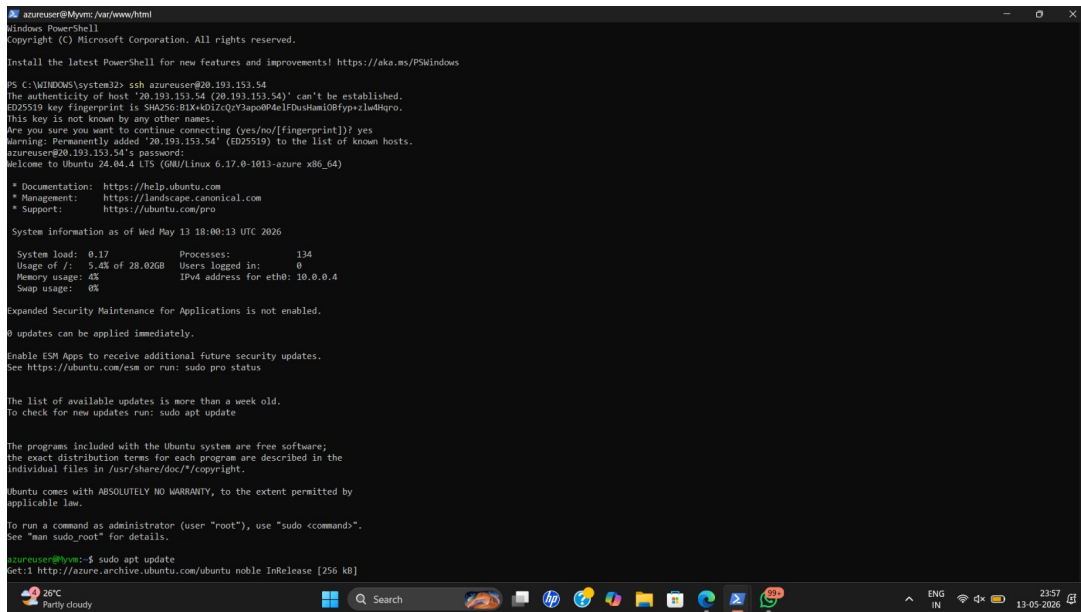


Figure 1: Azure Virtual Machine Dashboard

This screenshot shows the Azure portal where the Virtual Machine is successfully deployed. It displays the VM status, resource group, and public IP address assigned by Azure.

SSH Connection Through PowerShell



```
azureser@Myvm: /usr/www/html
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\WINDOWS\system32> ssh azureser@20.193.153.54
The authenticity of host '20.193.153.54 (20.193.153.54)' can't be established.
ED25519 key fingerprint is SHA256:R3X4ADZ2QzY3ap0P4e1F0u8amIG8Fypz1wBqro.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '20.193.153.54' (ED25519) to the list of known hosts.
azureser@20.193.153.54's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1013-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Wed May 13 18:00:13 UTC 2026

System load: 0.17               Processes: 134
Usage of /:  5.4% of 28.02GB     Users logged in: 0
Memory usage: 4%                IPv4 address for eth0: 10.0.0.4
Swap usage: 0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/*copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

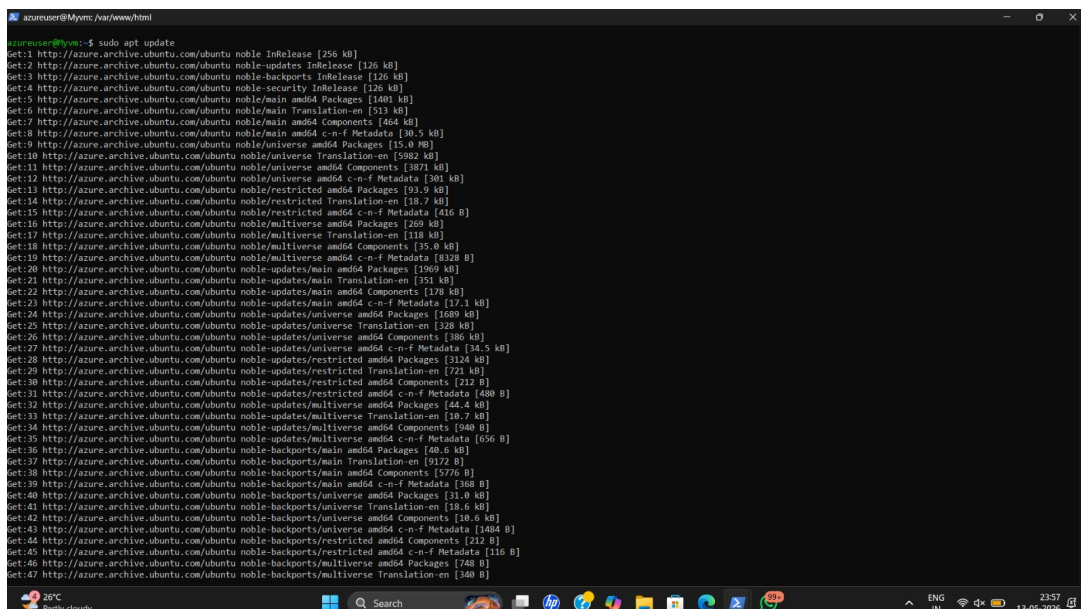
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

azureser@Myvm:~$ sudo apt update
Get:1 http://azure.archive.ubuntu.com/ubuntu noble InRelease [256 kB]
Get:2 http://azure.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://azure.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://azure.archive.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:5 http://azure.archive.ubuntu.com/ubuntu noble/main amd64 Packages [1400 kB]
Get:6 http://azure.archive.ubuntu.com/ubuntu noble/main Translation-en [513 kB]
Get:7 http://azure.archive.ubuntu.com/ubuntu noble/main amd64 Components [464 kB]
Get:8 http://azure.archive.ubuntu.com/ubuntu noble/main amd64 c-n-f Metadata [10.2 kB]
Get:9 http://azure.archive.ubuntu.com/ubuntu noble/universe amd64 Packages [15.0 MB]
Get:10 http://azure.archive.ubuntu.com/ubuntu noble/universe Translation-en [5982 kB]
Get:11 http://azure.archive.ubuntu.com/ubuntu noble/universe amd64 Components [387.1 kB]
Get:12 http://azure.archive.ubuntu.com/ubuntu noble/universe amd64 c-n-f Metadata [380 kB]
Get:13 http://azure.archive.ubuntu.com/ubuntu noble/restricted amd64 Packages [93.9 kB]
Get:14 http://azure.archive.ubuntu.com/ubuntu noble/restricted Translation-en [18.7 kB]
Get:15 http://azure.archive.ubuntu.com/ubuntu noble/restricted amd64 c-n-f Metadata [416 B]
Get:16 http://azure.archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages [269 kB]
Get:17 http://azure.archive.ubuntu.com/ubuntu noble/multiverse Translation-en [118 kB]
Get:18 http://azure.archive.ubuntu.com/ubuntu noble/multiverse amd64 Components [25.0 kB]
Get:19 http://azure.archive.ubuntu.com/ubuntu noble/multiverse amd64 c-n-f Metadata [8328 B]
Get:20 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1969 kB]
Get:21 http://azure.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [351 kB]
Get:22 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [119 kB]
Get:23 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [17.1 kB]
Get:24 http://azure.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1689 kB]
Get:25 http://azure.archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [328 kB]
Get:26 http://azure.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [386 kB]
Get:27 http://azure.archive.ubuntu.com/ubuntu noble-updates/universe amd64 c-n-f Metadata [34.5 kB]
Get:28 http://azure.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [124 kB]
Get:29 http://azure.archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [721 kB]
Get:30 http://azure.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:31 http://azure.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 c-n-f Metadata [480 B]
Get:32 http://azure.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Packages [44.4 kB]
Get:33 http://azure.archive.ubuntu.com/ubuntu noble-updates/multiverse Translation-en [10.7 kB]
Get:34 http://azure.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:35 http://azure.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 c-n-f Metadata [656 B]
Get:36 http://azure.archive.ubuntu.com/ubuntu noble-backports/main amd64 Packages [48.6 kB]
Get:37 http://azure.archive.ubuntu.com/ubuntu noble-backports/main Translation-en [9172 B]
Get:38 http://azure.archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [5776 B]
Get:39 http://azure.archive.ubuntu.com/ubuntu noble-backports/main amd64 c-n-f Metadata [368 B]
Get:40 http://azure.archive.ubuntu.com/ubuntu noble-backports/universe amd64 Packages [31.0 kB]
Get:41 http://azure.archive.ubuntu.com/ubuntu noble-backports/universe Translation-en [18.6 kB]
Get:42 http://azure.archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [10.6 kB]
Get:43 http://azure.archive.ubuntu.com/ubuntu noble-backports/universe amd64 c-n-f Metadata [1484 B]
Get:44 http://azure.archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [212 B]
Get:45 http://azure.archive.ubuntu.com/ubuntu noble-backports/restricted amd64 c-n-f Metadata [116 B]
Get:46 http://azure.archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Packages [748 B]
Get:47 http://azure.archive.ubuntu.com/ubuntu noble-backports/multiverse Translation-en [340 B]
```

Figure 2: SSH Connection to Virtual Machine

This image shows the successful SSH login into the Ubuntu server using PowerShell. The connection confirms that remote access to the cloud machine is working properly.

Updating Ubuntu Packages



```
azureser@Myvm:~$ sudo apt update
Get:1 http://azure.archive.ubuntu.com/ubuntu noble InRelease [256 kB]
Get:2 http://azure.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://azure.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://azure.archive.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:5 http://azure.archive.ubuntu.com/ubuntu noble/main amd64 Packages [1400 kB]
Get:6 http://azure.archive.ubuntu.com/ubuntu noble/main Translation-en [513 kB]
Get:7 http://azure.archive.ubuntu.com/ubuntu noble/main amd64 Components [464 kB]
Get:8 http://azure.archive.ubuntu.com/ubuntu noble/main amd64 c-n-f Metadata [10.2 kB]
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Get:10 http://azure.archive.ubuntu.com/ubuntu noble/universe Translation-en [5982 kB]
Get:11 http://azure.archive.ubuntu.com/ubuntu noble/universe amd64 Components [387.1 kB]
Get:12 http://azure.archive.ubuntu.com/ubuntu noble/universe amd64 c-n-f Metadata [380 kB]
Get:13 http://azure.archive.ubuntu.com/ubuntu noble/restricted amd64 Packages [93.9 kB]
Get:14 http://azure.archive.ubuntu.com/ubuntu noble/restricted Translation-en [18.7 kB]
Get:15 http://azure.archive.ubuntu.com/ubuntu noble/restricted amd64 c-n-f Metadata [416 B]
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Get:20 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1969 kB]
Get:21 http://azure.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [351 kB]
Get:22 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [119 kB]
Get:23 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [17.1 kB]
Get:24 http://azure.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1689 kB]
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Get:27 http://azure.archive.ubuntu.com/ubuntu noble-updates/universe amd64 c-n-f Metadata [34.5 kB]
Get:28 http://azure.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [124 kB]
Get:29 http://azure.archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [721 kB]
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Get:31 http://azure.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 c-n-f Metadata [480 B]
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Get:37 http://azure.archive.ubuntu.com/ubuntu noble-backports/main Translation-en [9172 B]
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Get:42 http://azure.archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [10.6 kB]
Get:43 http://azure.archive.ubuntu.com/ubuntu noble-backports/universe amd64 c-n-f Metadata [1484 B]
Get:44 http://azure.archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [212 B]
Get:45 http://azure.archive.ubuntu.com/ubuntu noble-backports/restricted amd64 c-n-f Metadata [116 B]
Get:46 http://azure.archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Packages [748 B]
Get:47 http://azure.archive.ubuntu.com/ubuntu noble-backports/multiverse Translation-en [340 B]
```

Figure 3: System Update Process

This screenshot shows the execution of the `sudo apt update` command. It updates the package list in the Ubuntu system to ensure the latest software versions are available for installation.

Apache2 Web Server Installation

```

azuser@Myvm: /var/www/html
get:64 http://azure.archive.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [288 B]
get:65 http://azure.archive.ubuntu.com/ubuntu noble-security/multiverse amd64 c-n-f Metadata [396 B]
Fetched 44.9 MB in 7s (6329 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
3 packages can be upgraded. Run 'apt list --upgradable' to see them.
azuser@Myvm:~$ sudo apt install apache2-y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
E: Unable to locate package apache2-y
azuser@Myvm:~$ sudo apt install apache2 -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  apache2-bin apache2-data apache2-utils libapr1t64 libaprutil1-dbd-sqlite3 libaprutil1-ldap libaprutil1t64
  liblua5.4-0 ssl-cert
Suggested packages:
  apache2-doc apache2-suexec-pristine | apache2-suexec-custom www-browser
The following NEW packages will be installed:
  apache2 apache2-bin apache2-data apache2-utils libapr1t64 libaprutil1-dbd-sqlite3 libaprutil1-ldap libaprutil1t64
  liblua5.4-0 ssl-cert
0 upgraded, 10 newly installed, 0 to remove and 8 not upgraded.
Need to get 2092 kB of archives.
After this operation, 8117 kB of additional disk space will be used.
Get:1 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 libapr1t64 amd64 1.7.2-3.1ubuntu0.1 [180 kB]
Get:2 http://azure.archive.ubuntu.com/ubuntu noble/main amd64 libaprutil1t64 amd64 1.6.3-1.1ubuntu7 [191.9 kB]
Get:3 http://azure.archive.ubuntu.com/ubuntu noble/main amd64 libaprutil1-dbd-sqlite3 amd64 1.6.3-1.1ubuntu7 [111.2 kB]
Get:4 http://azure.archive.ubuntu.com/ubuntu noble/main amd64 libaprutil1-ldap amd64 1.6.3-1.1ubuntu7 [9116 B]
Get:5 http://azure.archive.ubuntu.com/ubuntu noble/main amd64 liblua5.4-0 amd64 5.4.6-2build2 [166 kB]
Get:6 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2-bin amd64 2.4.58-1ubuntu8.12 [1336 kB]
Get:7 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2-data all 2.4.58-1ubuntu8.12 [163 kB]
Get:8 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2-utils amd64 2.4.58-1ubuntu8.12 [190.9 kB]
Get:9 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2 amd64 2.4.58-1ubuntu8.12 [190.2 kB]
Get:10 http://azure.archive.ubuntu.com/ubuntu noble/main amd64 ssl-cert all 1.1.2ubuntu1 [17.8 kB]
Fetched 2092 kB in 1s (2532 kB/s)
Preconfiguring packages ...
Selecting previously unselected package libapr1t64:amd64.
(Reading database ... 69211 files and directories currently installed.)
Preparing to unpack .../0-libapr1t64_1.7.2-3.1ubuntu0.1_amd64.deb ...
Unpacking libapr1t64:amd64 (1.7.2-3.1ubuntu0.1) ...
Selecting previously unselected package libaprutil1t64:amd64.
Preparing to unpack .../1-libaprutil1t64_1.6.3-1.1ubuntu7_amd64.deb ...
Unpacking libaprutil1t64:amd64 (1.6.3-1.1ubuntu7) ...
Selecting previously unselected package libaprutil1-dbd-sqlite3:amd64.
Preparing to unpack .../2-libaprutil1-dbd-sqlite3_1.6.3-1.1ubuntu7_amd64.deb ...
Unpacking libaprutil1-dbd-sqlite3:amd64 (1.6.3-1.1ubuntu7) ...

```

Figure 4: Apache Web Server Installation

This image shows the installation of Apache2 web server using the Ubuntu package manager. Apache is required to host and serve the static website over HTTP.

Website File Creation in Apache Directory

```

azuser@Myvm: /var/www/html
Setting up apache2-bin (2.4.58-1ubuntu8.12) ...
Setting up apache2 (2.4.58-1ubuntu8.12) ...
Enabling module mpm_event.
Enabling module mpm_core.
Enabling module authz_host.
Enabling module authn_core.
Enabling module auth_basic.
Enabling module access_compat.
Enabling module authn_file.
Enabling module auth_user.
Enabling module alias.
Enabling module dir.
Enabling module autoindex.
Enabling module env.
Enabling module mime.
Enabling module negotiation.
Enabling module setenvif.
Enabling module filter.
Enabling module deflate.
Enabling module status.
Enabling module reqtimeout.
Enabling conf charset.
Enabling conf localized-error-pages.
Enabling conf other-vhosts-access-log.
Enabling conf security.
Enabling conf serve-cgi-bin.
Enabling site 000-default.
Created symlink /etc/systemd/system/multi-user.target.wants/apache2.service → /usr/lib/systemd/system/apache2.service.
Created symlink /etc/systemd/system/multi-user.target.wants/apache-htcacheclean.service → /usr/lib/systemd/system/apache-htcacheclean.service.
Processing triggers for ufw (0.36-2) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.7) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
azuser@Myvm:~$ cd /var/www/html
azuser@Myvm: /var/www/html$ sudo rm index.html
azuser@Myvm: /var/www/html$ sudo nano index.html

```

Figure 5: Creating Website File

This screenshot shows the creation of the index.html file inside the Apache root directory. This file contains the website content that will be displayed in the browser when accessing the public IP address.

Final Hosted Website Output

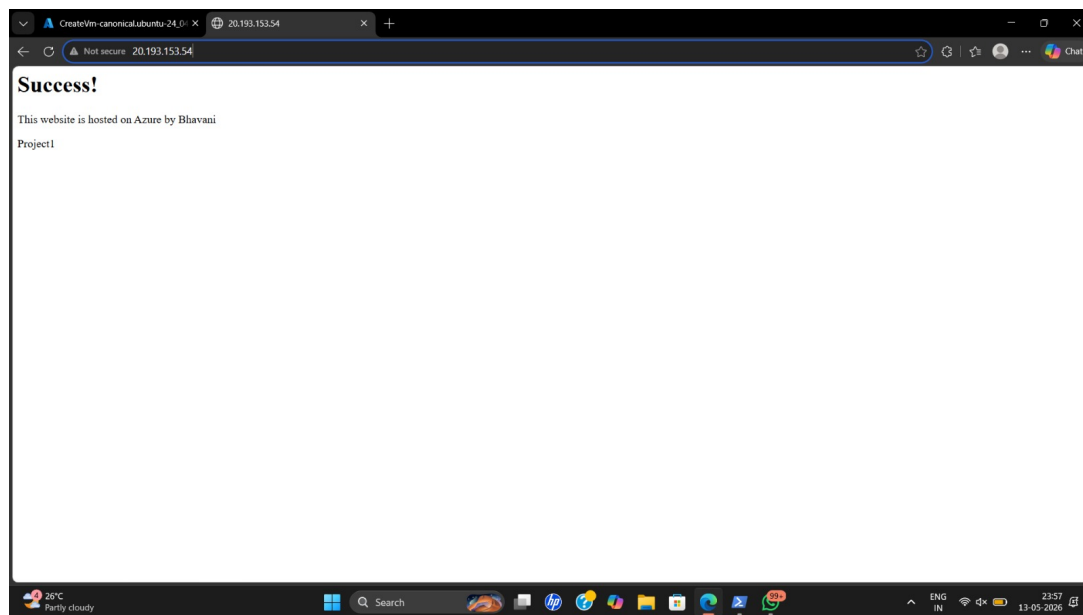


Figure 6: Hosted Website Running on Azure VM

This screenshot shows the final output of the project where the static website is successfully hosted on the Azure Virtual Machine. The webpage is accessed using the public IP address of the VM through a web browser.

The page confirms that the Apache Web Server is working correctly and the HTML file placed inside the `/var/www/html` directory is being served successfully. This proves that the cloud deployment was completed without errors.

COST ANALYSIS

Microsoft Azure follows a pay-as-you-go pricing model. For educational and small-scale projects, costs can be minimized by selecting low-resource virtual machines and stopping unused resources.

The major cost factors include:

- Virtual Machine Compute Cost
- Storage Cost
- Public IP Allocation
- Network Usage

ADVANTAGES

- Easy website deployment
- Low infrastructure cost
- Better scalability
- Remote server access
- Improved reliability
- Easy maintenance
- High availability

PERFORMANCE BENEFITS

Azure provides high availability and reliable performance through globally distributed data centers. The hosted website benefits from:

- Fast deployment
- High uptime
- Global accessibility
- Scalability on demand
- Efficient resource utilization

COMPARISON WITH TRADITIONAL HOSTING

Feature	Traditional Hosting	Azure Cloud Hosting
Initial Cost	High	Low
Scalability	Limited	High
Maintenance	Manual	Managed
Availability	Depends on Hardware	High Availability
Deployment Speed	Slow	Fast

APPLICATIONS

Applications of Azure Virtual Machine hosting include:

- Educational Projects
- Website Hosting
- Startup Deployment
- Application Hosting
- Cloud Computing Training
- Linux Server Administration

RESULTS AND DISCUSSION

The Azure Virtual Machine was successfully created and configured using Microsoft Azure cloud services. Necessary networking components such as the Virtual Network, Public IP Address, and Network Security Group were configured to enable secure communication between users and the server. The Ubuntu 22.04 LTS operating system was deployed successfully and remote access was established through SSH.

Apache HTTP Server was installed and configured on the virtual machine. A static website developed using HTML and CSS was placed in the Apache web directory and hosted successfully. The website was accessible through the public IP address assigned to the Azure Virtual Machine, confirming that the web server and networking configurations were functioning correctly.

The project demonstrated the practical implementation of cloud computing concepts, including Infrastructure as a Service (IaaS), virtual machine deployment, server administration, and web hosting. The successful deployment verified that Azure provides a reliable platform for hosting web applications while reducing the need for physical infrastructure.

The hosted website showed stable performance and could be accessed remotely through a standard web browser. The implementation also provided hands-on experience in Linux system management, package installation, firewall configuration, and cloud resource management.

Overall, all project objectives were achieved successfully. The project enhanced understanding of cloud technologies and provided practical exposure to real-world website deployment and server management processes. The knowledge gained through this implementation can be applied to more advanced cloud-based applications and services in the future.

FUTURE SCOPE

Future improvements that can be implemented include:

- Hosting dynamic websites
- Deploying databases
- Using HTTPS and SSL certificates
- Configuring custom domain names
- Implementing load balancing
- Automating deployment pipelines
- Using Docker containers

CONCLUSION

This project successfully demonstrated the creation of a Linux Virtual Machine on Microsoft Azure and hosting of a static website using Apache Web Server. The implementation provided practical understanding of cloud computing concepts, Linux server administration, networking configuration, and website deployment.

The website was successfully hosted and verified through the public IP address of the Azure Virtual Machine. The project also helped in understanding real-world cloud infrastructure management and server deployment processes.

Overall, the project provided valuable hands-on experience in Microsoft Azure cloud services and web hosting technologies.

This project also helped in developing practical problem-solving skills, especially while handling real-world issues such as connectivity errors and service configuration problems. It provided a better understanding of how cloud systems operate behind the scenes.

Overall, the experience gained from this project is highly valuable for future work in cloud computing, DevOps, and system administration domains.

LEARNING OUTCOMES

After completing this project, the following skills and knowledge were gained:

- **Understanding of Cloud Computing Architecture:** Learned how cloud resources such as virtual machines, networking services, and storage components work together to provide scalable computing infrastructure.
- **Hands-on Experience with Microsoft Azure:** Gained practical experience in creating, configuring, and managing cloud resources using the Microsoft Azure platform.
- **Linux Server Management Skills:** Developed basic Linux administration skills including package management, file handling, and server configuration.
- **Web Server Installation and Configuration:** Learned how to install, configure, and manage the Apache HTTP Server for hosting web applications.
- **Networking and Firewall Configuration:** Understood the importance of network security and configured inbound rules to allow secure access to the hosted website.
- **SSH Remote Server Access:** Learned how to securely connect to and manage remote cloud servers using SSH protocols.
- **Static Website Deployment:** Acquired practical knowledge of deploying HTML and CSS websites on a cloud-based server environment.

REAL-WORLD EXPERIENCE AND LEARNING INSIGHTS

During the execution of this project, a clear understanding of how real cloud infrastructure operates was developed. Unlike traditional local systems, cloud platforms like Microsoft Azure require proper configuration of multiple components such as networking, security rules, and virtual machines before any application can be deployed successfully.

One of the most important learnings from this project was understanding how different cloud services work together. The virtual machine acts as the computing unit, the network security group controls traffic flow, and the public IP address enables external access. When all these components are correctly configured, they collectively create a fully functional cloud-hosted environment.

Another important insight gained was the importance of troubleshooting in real systems. While working on SSH connection and Apache server setup, small configuration issues such as port blocking or service not running can prevent the system from working. Identifying and resolving these issues improved practical problem-solving skills.

This project also provided exposure to Linux-based server management, including file handling, package installation, and service control. These skills are widely used in DevOps and cloud engineering roles in the industry.

Overall, the project helped in understanding how theoretical cloud computing concepts are implemented in real-world scenarios and how professionals manage cloud infrastructure in production environments.

PROJECT IMPACT

The successful completion of this project demonstrates how cloud computing can simplify website deployment and infrastructure management. By utilizing Microsoft Azure, the need for physical hardware was eliminated while maintaining accessibility, scalability, and reliability.

The project highlights the importance of cloud technologies in modern IT environments and showcases how organizations can efficiently deploy and manage web applications using virtualized infrastructure.

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